A Low-Cost Semi-Autonomous Rover for Laser Weed Control with Smartphone-Based Field Mapping

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Abstract—Herbicide-resistant weeds, residue regulations, and the rising cost of manual weeding create pressure for non-chemical weed control on smallholder farms, where commercial laser-weeding platforms are prohibitively expensive. This paper presents a low-cost semi-autonomous weeding rover that plans missions from a single smartphone photograph: the operator taps the four corners of a field block in the app, and the rover's embedded computer warps the photo into a metric top-down (bird's-eye-view) image, segments vegetation using an excess-green threshold with Otsu-backed automation, extracts crop rows by column projection, regularizes them against the operator-declared row count, and emits a boustrophedon waypoint mission that a 3-DOF targeting arm follows to aim the laser. The perception step is quantitatively evaluated on a reproducible synthetic benchmark spanning camera tilt, row pitch, and corner-tapping error. Regularized row extraction restored exact row-count detection in 100 % of trials at tilts up to 60°, whereas the unregularized pipeline dropped to 75 % at 50–60° on a tight 0.6 m row pitch. Row position error remained below 8 cm at 60°, and the complete real-photo pipeline ran in 19.5 ms per field block. All software stages are validated end-to-end in a physics-visualized 3-D tandem simulation, and the system is embodied in a working chassis prototype with a full KiCad electrical design.

Index Terms—weeding robot, laser weeding, precision agriculture, bird's-eye view, row detection, low-cost robotics, embedded vision

I. Introduction

Weeds are a primary cause of crop-yield loss, and their control still relies heavily on manual labor or herbicide programs whose residue and resistance drawbacks are increasingly difficult to sustain [1]. Laser weeding is an attractive non-chemical alternative: a focused beam damages weed meristems within a fraction of a second, leaves no residue, and requires no contact with the crop. However, reported systems—academic platforms built on high-end field robots [3], [4] and commercial units such as the Carbon LaserWeeder [8]—target large farms and remain far outside the budget envelope of smallholder agriculture.

Between these poles sits a gap: semi-autonomous rovers that borrow human judgment where it is cheap (one photograph per field block, taken with a commodity smartphone) and automate what is expensive or dangerous (navigation, repeatability, laser aiming). Prior vision-based weeding robots demonstrated row tracking from an on-board camera [2], but typically assume continuous in-motion perception. Our requirements differ: targeted laser work needs an accurate global lane map before the rover enters the field, and the only sensor guaranteed to be available to a smallholder is a phone camera.

This paper makes the following contributions:

1) A complete, low-cost architecture for laser weeding in which a smartphone photo plus four corner taps is the sole mission-autonomy input, processed on a Raspberry Pi-class single-board computer (Section II).
2) A field-perception pipeline that is deliberately robust to arbitrary handheld camera pose, built around a corner-driven bird's-eye-view (BEV) warp and an expected-count row regularizer, with a quantitative account of why tilts degrade raw projection and how regularization removes that failure mode (Sections III and V).
3) A reproducible synthetic-field robustness benchmark (camera tilt, row pitch, corner-tap noise) and a real-photo timing characterization, released as runnable artifacts with the system (Section V).
4) An end-to-end 3-D tandem simulation whose replay drives the same waypoints and zap decisions the hardware executes, plus a physical chassis prototype and a connectivity-verified KiCad electrical design (Sections II-B and IV).

II. System Architecture

A. Hardware Platform

The rover chassis is a four-wheel differentially driven platform carrying an RPi-class single-board computer that hosts the mission server, an ESP32 acting as the real-time motor bridge, and an ultrasonic proximity sensor for headland safety stop. Above the chassis sits a 3-DOF arm: two continuous-axis servos orient a laser diode in pan and tilt, and the third servo deploys/masks the beam through a gimbal-mounted pointer. The complete electrical system is described in Section II-B; it was machine-generated

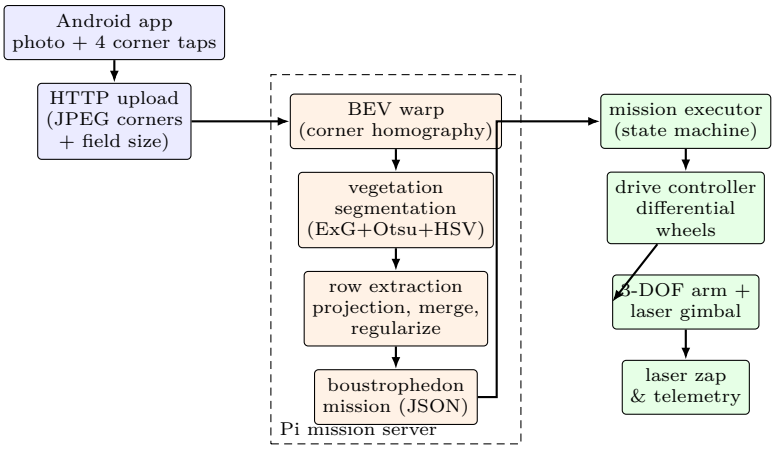


Fig. 1. System architecture. The phone contributes exactly one photograph and four corner taps; all perception, planning, and execution happens on the rover's on-board computer.

as a fully-wired KiCad sheet and gated on netlist/ERC verification.

B. Electrical Design

The complete electrical system was drawn in KiCad 9 as a generated, fully-wired single sheet: a 3S LiPo feeds a main switch and 15 A fuse into a shunt-monitored 12 V rail (INA219 over I^2C), a 5 V buck powers the Pi-class SBC, a dedicated 6 V UBEC rail powers the arm servos in isolation, dual MD13S bridges drive the wheel groups, and the laser line runs through a hardware key-lock interlock that the on-board computer's laser-enable signal must logically AND with before the driver stage fires. Two HC-SR04 ultrasonics and two VL53L0X ToF sensors provide front and rear proximity sensing, and an emergency-stop button breaks the laser enable and motor command chain. Every pin of every component bears its net label, so net connectivity is anchored by labels even where a route is drawn only logically; the exported netlist reproduces all 37 named nets exactly as designed—this was not the case for an earlier revision, whose routing pass allowed different nets to share wire cells and thereby merged most of the sheet into a single net, a failure mode caught exactly by this verification gate and fixed by claiming cells exclusively during routing. ERC on the fixed sheet reports zero connectivity violations (the remaining notices concern the symbol-library table, a configuration matter). Fig. 2 shows the result; the KiCad source, PDF export, and netlist ship with the repository artifacts.

C. Software Organization

The pipeline is deliberately split at exactly one boundary: a JSON/HTTP exchange between the phone app and the rover. The app uploads a JPEG image, the four tapped corners in photo coordinates, and the field's physical dimensions (width, length, and the operator's row count). Everything downstream—warping, segmentation,

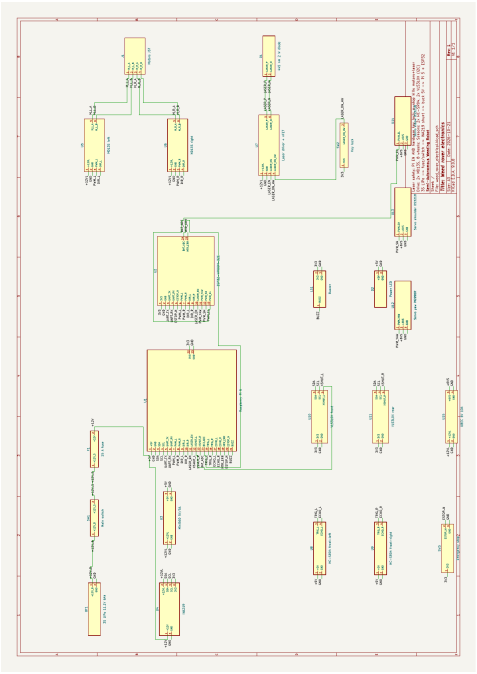


Fig. 2. Complete KiCad electrical schematic of the rover. Net labels at every pin keep connectivity legible; netlist export and ERC serve as the verification gate.

row extraction, mission planning, execution, and laser control—lives on the rover, so perception parameters can be retuned per crop without shipping a new phone application.

III. Field Perception Pipeline

A. Corner-Driven BEV Warp

Rather than requiring a nadir (top-down) photograph, the pipeline accepts any handheld framing: the four tapped corner points, ordered top-left, top-right, bottom-right, bottom-left, define a field quadrilateral in the photo, which is warped with a single projective transform into a BEV image of 900 px width scaled to the field's real aspect ratio. Corner-driven homography recovers the plane geometry exactly, so the geometric tolerance of the method reproduces the skill of a human operator; robustness to camera tilt is bounded not by geometry but by the image-quality consequences of tilt—the remaining sections show that the perception stages, not the warp, bear that burden.

B. Vegetation Segmentation

Vegetation is segmented using the classical excess-green quantity $E_xG = 2g - r - b$ [6], whose threshold is chosen automatically by Otsu's method [7] on a contrast-scaled version of the index, then intersected with an HSV green-dominance test and subjected to morphological opening/closing to stabilize against soil noise. On the real field photograph used in Section V, this stage is both the most expensive and the most decision-dense step: it



Fig. 3. Real-field pipeline demonstration: (a) smartphone photo of a seedling block, (b) metric BEV after corner warp, (c) detected rows overlaid (yellow). 7/7 rows on the tapped block; 19.5 ms end-to-end.

fixes the vegetation ratio that anchors all downstream row reasoning.

C. Row Extraction with Expected-Count Regularization

Rows are extracted from the BEV's column-wise projection of the vegetation mask. Segments above 25 % of the projection maximum are candidate rows; short segments are merged when their gap is below 0.6× the expected row pitch, and their centers are the raw row estimates. Raw projection is fragile for two reasons: photographs taken at an oblique angle compress the far edge of the field, spreading blur and JPEG artifacts over several centimeters of ground sampling distance, and thin row bands can dip below the projection threshold in gaps between plants and reappear after merging. The regularizer exploits the one piece of agronomic knowledge nobody on-site lacks: the crop was planted with an approximately constant row pitch, and usually a row count the operator knows. Given the expected count k, the regularizer snaps the row centers to a uniform k-slot grid that best matches the detected set—anchored in the median row spacing—so a row merged into a neighbor or lost to a gap is restored symmetrically around the failed band rather than left out.

D. Worked Example on a Real Photograph

Fig. 3 shows the pipeline on an authentic field photo: (a) the operator-visible scene, (b) the BEV computed from the block corners, and (c) the final row overlay: 7 of 7 rows detected, as discussed quantitatively in Section V.

Fig. 4 shows the synthetic-field equivalent at an extreme 60° camera tilt with 0.6 m row pitch: the raw merged row list dissolves lost rows that the regularizer recovers.

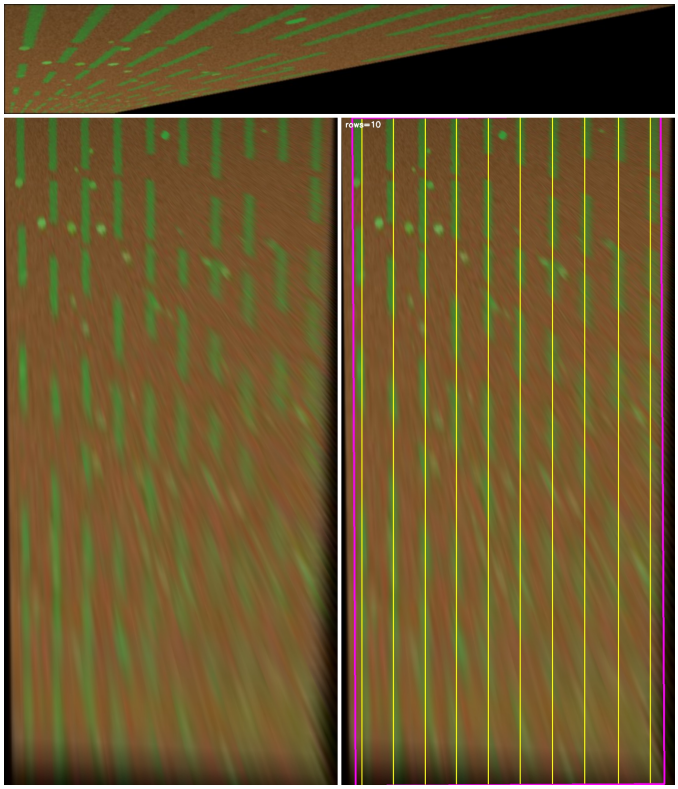


Fig. 4. Synthetic benchmark example at 60° tilt, 0.6 m pitch: (a) rendered tilted photo, (b) metric BEV, (c) final overlay. Row gaps and weed clutter make raw projection fragile; regularization restores the operator-declared row layout.

IV. Mission Planning and Execution

A. Boustrophedon Planning

Row centers and the field metric dimensions yield a lane-following mission: each row lane is traversed end-to-end with alternating directions, inset by a 0.2 m headland margin at both ends (one waypoint per lane end, so a k-row block produces $2k$ waypoints), emitted as plain JSON. Keeping the mission language JSON and shallow means the same artifact drives the 3-D twin simulation, the phone app's preview, and the on-board executor without translation. Fig. 5 shows an authentic mission-server output for a real user upload: nine lane rows and the boustrophedon traversal bounds computed from the tapped corners.

B. Following, Aiming, Zapping

At run time the executor interleaves lane travel and a head-turn-zap pattern: the rover drives to waypoints with a differential-wheel controller routed through the ESP32 (which averages odometry and watches the ultrasonic stop-sensor), suspends lane travel at each pointing station, aims the arm using the lane-relative weed coordinates computed from the row map, gates zap duration to weed width, and remounts. A finite-state machine arbitrates modes so that perception, navigation, and laser safety never co-drive the same actuator.

C. 3-D Simulation Validation

Before the hardware ever reached the field, complete missions were replayed in a Blender-rendered 3-D twin of the rover in which the same mission JSON drives the same state machine interfacing to a simulated chassis/arm/laser. Fig. 6 shows four frames of a mission replay spanning the entrance, mid-lane travel, a zap event, and a row-end turn. Because the simulation consumes the production mission artifact and state machine, it validates the whole chain from map JSON onward.



Fig. 5. Authentic mission-server output for a real field upload: detected lane rows (yellow) and the traversal bounds (magenta) computed from one tapped-corner photo upload.

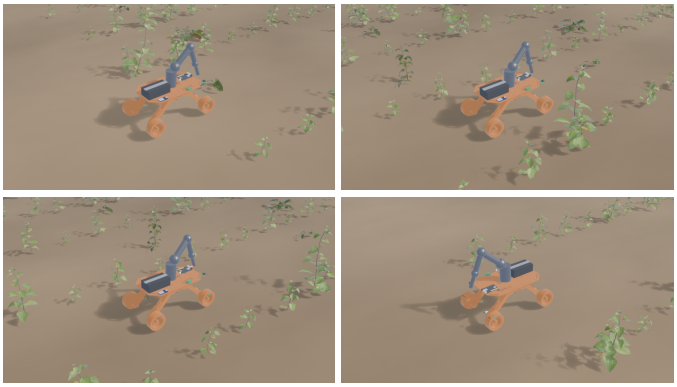


Fig. 6. 3-D tandem simulation of the physical mission replay: field entrance (top left), mid-lane travel (top right), laser zap engagement (bottom left), and end-of-mission (bottom right).

V. Evaluation

A. Benchmark Design

Because no instrumented field trials exist yet, perception robustness is quantified on a released synthetic benchmark with ground truth (Navigation/benchmarks/). Fields are generated at 6 m × 12 m with 100 px m^{-1}, containing thin (0.15 m) crop bands broken into plant-run segments with realistic gaps, 90 scattered weed blobs, pixel grain, low-frequency soil mottling, and per-trial random seeding. A pinhole camera model renders each field as if photographed by a tilted phone: the plane is rotated about the horizontal axis by angle θ, projected from 2 m above the near edge, then given a lighting gradient, mild Gaussian blur, JPEG compression (quality 85), and Gaussian corner-tap error ($\sigma = 3$ px) added to the four projected corners, which are passed to the production pipeline as the operator's taps. Each configuration is run over 8 random seeds.

TABLE I
Tilt-sweep exact row-count rate (%) over 8 seeds per cell.

		1.2 m pitch		0.6 m pitch	
Arm	θ	rate	$\lvert\Delta x\rvert$ cm	rate	$\lvert\Delta x\rvert$ cm
raw	0°	100	2.1	100	1.8
raw	30°	100	5.4	100	4.7
raw	50°	100	6.1	75	6.9
raw	60°	100	7.5	75	6.8
reg	0°	100	2.1	100	1.8
reg	30°	100	5.4	100	4.7
reg	50°	100	6.1	100	6.9
reg	60°	100	7.5	100	6.9

Two sets are reported: a tilt sweep ($\theta \in [0°..60°]$ step 10°, both 1.2 m and 0.6 m row pitches, i.e. 2 arms ×14 settings) and a tap sweep ($\sigma \in [0, 2, 4, 8, 16]$ px at 30°). An ablation arm disables only the regularizer, keeping merge and thresholds identical. The complete sweep comprises 192 pipeline runs. Additional realism stressors are deliberately modest; no claim is made that synthetic trials substitute for field validation.

Metrics: row-count error (detected minus expected), exact-count rate, mean row position error when the count is exact, and mission waypoint-count error. All timing is measured on an x86-64 laptop (Intel i3-1215U), reported as a platform-relative reference—the production host is the rover's single-board computer.

B. Tilt Robustness

Table I and Fig. 7 show the central result. At 1.2 m pitch, both arms stay at 100 % exact counts at every tilt; the wide pitch leaves so much projection headroom that even 60° cannot collapse neighboring peaks. At 0.6 m pitch, the unregularized pipeline merged or lost rows in 2 of 8 seeds at both 50° and 60° (mean count error −0.25 and −0.375), while the regularized pipeline detected the exact count in every trial. Two camera-physical mechanisms explain the raw curve's shape: far-edge ground-sampling-distance inflation (the ratio of near-edge to far-edge ground resolution reaches ≈ 6× at 60° for a 12 m field from 2 m above the near edge), so blur smears adjacent band projections together, and gap-driven band thinning, which drops whole rows below the segment threshold. Regularization repairs both failure modes because a lost/merged row leaves a predictable hole in a near-uniform grid.

Row position error grows monotonically with tilt in both arms (from ≈ 2 cm level at 0° to 7.5 cm at 60°) because the same GSD inflation perturbs the mode-center estimate of the far rows; it never threatens mission validity (bold in

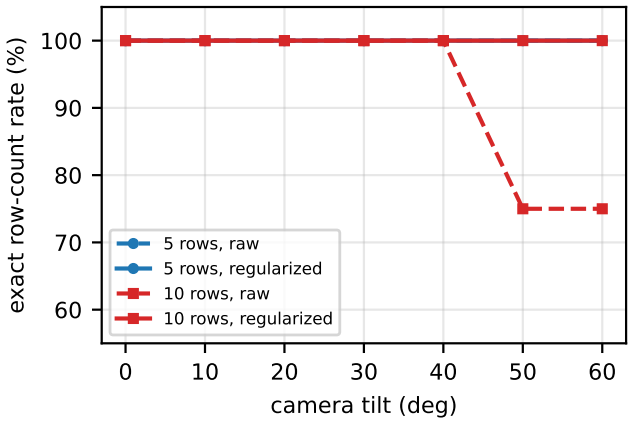


Fig. 7. Exact row-count rate versus camera tilt for both row pitches and pipeline variants. The unregularized pipeline (dashed) collapses at 50–60° on the 0.6 m pitch; regularization (solid) stays at 100 % throughout.

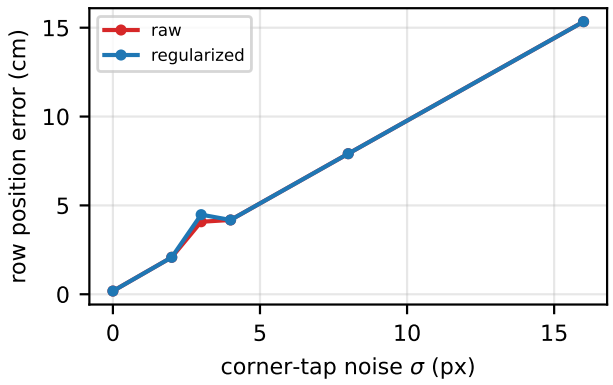


Fig. 8. Row position error versus corner-tap noise at 30° tilt (1.2 m pitch). Both arms linearly track tap error; the arms coincide because tap noise perturbs position, not count, on this pitch.

Table I), and waypoint-count errors were zero in all trials where the count was exact, and exactly ±1 lane end per lost row in the raw arm's failed cells.

C. Corner-Tap Robustness

Fig. 8 varies the Gaussian corner-tap error at fixed 30° tilt. Row position error grows essentially linearly with tap error—from 0.2 cm at $\sigma = 0$ to 15.3 cm at $\sigma = 16$ px—while row counts and mission validity remain perfect at both 1.2 m and 0.6 m pitch. The practical reading: casual tapping (a few pixels) costs a few centimeters of lane alignment, which is far below the row-following tolerance of a laser pointer of beam width at arm reach.

D. Timing on a Real Photograph

Table II reports end-to-end cost on the real photograph of Fig. 3 (tapped block: 1580×1229 px → BEV 900 × 694 px; minimum of 7 repeats). Segmentation dominates once again because it is the only stage processing full color at full resolution; the projection/regularization chain is negligible (< 2 ms). The full-frame photo (both blocks plus surroundings, 2816×1536 px) processed at 88 ms end-to-end, confirming that the sensible strategy of tapping the field block rather than the whole frame keeps the pipeline well under real-time on the same class of CPU.

TABLE II
Real-photo pipeline timing (minimum of 7 repeats, i3-1215U).

Stage	Time (ms)
BEV warp	1.2
vegetation segmentation	7.1
projection / merge / regularize	0.3
end-to-end (incl. image I/O)	19.5

VI. Limitations

Three limitations bound the current claims honestly. First, robustness numbers come from a synthetic benchmark with deliberately modest realism stressors, not from field trials; the synthetic generator and seeds are released precisely so that future in-field measurements can be compared like-for-like. Second, the regularizer trusts the operator's expected row count; a mis-typed count would snap rows into a wrong (but self-consistent) grid, which is why the interface polls it explicitly rather than inferring it. Third, the arm's aiming loop uses the pre-mission row map rather than per-frame closed-loop vision, so lane-keeping errors during travel add directly to aim error at resumption; a wrist camera would remove that coupling.

VII. Conclusion

A complete laser-weeding rover was presented whose entire mission-autonomy input is one smartphone photograph and four corner taps, processed entirely on a single-board computer in milliseconds. The central perception extension—an expected-count row regularizer against a corner-driven BEV warp—was shown to restore 100 % exact row detection where the raw pipeline falls to 75 % under 50–60° tilt at a realistic 0.6 m row pitch, and the system was validated end-to-end in a 3-D tandem simulation using the same production mission artifacts and state machine as the hardware, plus a working chassis prototype and a connectivity-clean KiCad electrical design. The natural next steps are instrumented field trials on the synthetic benchmark's conditions, weed-level (not row-level) targeting with per-stem aim, and an on-board wrist camera for closed-loop aiming between lane stations.

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